A Tensor-Based Expansion of the Cube Rule for Generative Nutrient Morphology

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Abstract

We introduce a tensor-based formulation of the Cube Rule of Food in which dishes are embedded in a third-order tensor $T \in \{0,1\}^{I \times J \times K}$ spanning cube morphology (index i), protein taxonomy (j), and starch substrate (k). This formulation supports reproducible placement of canonical Cube Rule foods, detection of collisions among distinct dishes at the same cell, and constrained generation of candidate dishes for vacant cells (i, j, k) with $T_{i,j,k} = 0$. To operationalize the framework, we use a constrained generative procedure that proposes candidate dishes for empty cells, enforces axis-consistency constraints on i, j, and k, and filters the results before downstream review.

1 Introduction

This project studies a formal extension of the Cube Rule of Food using a rank-3 tensor T over three categorical axes. We work with two *theoretical* state spaces: $\mathbb{F}_{\text{real}}$ (foods taken to have been cooked in recorded history—a set we approximate, not observe in full) and $\mathbb{F}_{\infty}$ (all conceivable foods, including in principle those not yet or never prepared—a space we cannot directly verify), with $\mathbb{F}_{\text{real}} \subseteq \mathbb{F}_{\infty}$. The immediate objective is to define a compact ontology in which each food in $\mathbb{F}_{\infty}$ is assigned a coordinate triple (i, j, k) in the discrete latent space $\{1, \ldots, I\} \times \{1, \ldots, J\} \times \{1, \ldots, K\}$, so that canonical Cube Rule examples are representable and ambiguous cases are explicit. Our framing treats occupancy, sparsity, and ambiguity as geometric properties of T rather than anecdotal judgments. Under this formulation, cuisine is represented as a discrete structure over the index space rather than as an informal list of examples.

2 Related Work

This work builds on the Cube Rule as a prior food-classification framework and introduces two external strands of literature to formalize the proposed tensor-based approach. The first strand is modern LLM control and prompting: prompt taxonomies, schema-constrained generation, grammar-constrained decoding, guided decoding, and retrieval-augmented generation provide language for the prompt design, structured decoding, exemplar use, and post-generation filtering described in the Methods section. The second strand is tensor and geometry-oriented representation theory: tensor completion, manifold hypothesis, simplicial complex constructions, persistent homology, and topological data analysis provide language for discussing empty-cell completion, clustered occupancy structure, and topological features of the support set $\mathcal{S} = \{(i, j, k) : T_{i,j,k} = 1\}$ in higher-order categorical space. Classical combinatorics references are also relevant for occupancy guarantees, including direct pigeonhole-style collision arguments when the number of foods exceeds

$|I \times J \times K| = IJK$. Taken together, these literatures provide the methodological and conceptual vocabulary used to define, populate, and analyze the proposed taxonomy.

3 Methods

We define two state spaces in direct analogy to the real numbers $\mathbb{R}$ (the space of all real numbers). Let $\mathbb{F}_{\text{real}}$ denote the *theoretical* set of all *real* foods: those taken to have been cooked and recorded in history (the culinary analogue of "real" in $\mathbb{R}$); we approximate this set with our catalog rather than observe it in full. Let $\mathbb{F}_\infty$ denote the *theoretical* set of all *conceivable* foods, including in principle dishes that have never been prepared—the full state space of possible foods, which we cannot directly verify—with $\mathbb{F}_{\text{real}} \subseteq \mathbb{F}_\infty$. Our tensor formalism maps elements of $\mathbb{F}_\infty$ to coordinates; the catalog we maintain is a finite subset of $\mathbb{F}_{\text{real}}$, and we speculate over elements of $\mathbb{F}_\infty \setminus \mathbb{F}_{\text{real}}$ for empty cells.

Let $T \in \{0,1\}^{I \times J \times K}$ denote a tensor indexed by cube type (axis $i \in \{1, \ldots, I\}$), protein class (axis $j \in \{1, \ldots, J\}$), and starch type (axis $k \in \{1, \ldots, K\}$). The starch axis includes a distinguished `none` value for dishes with no structural starch substrate. We set $T_{i,j,k} = 1$ when at least one food maps to index triple (i, j, k); the *support set* is $\mathcal{S} = \{(i, j, k) : T_{i,j,k} = 1\}$ and the set of *empty cells* is $\mathcal{E} = \{(i, j, k) : T_{i,j,k} = 0\}$. In implementation, each occupied cell may contain multiple candidate foods, so the data pipeline stores per-food records and derives occupancy from those records; formally we retain binary $T_{i,j,k}$ while allowing multiple foods per (i, j, k) in the database. This can be interpreted as a categorical incidence tensor with sparse $|\mathcal{S}| \ll IJK$. Operationally, this turns informal culinary intuition into an explicit indexed representation.

Under the Cube Rule, salad morphology means no structural starch faces. In the tensor, such cases occupy the slice $(i_{\text{salad}}, j, k_{\text{none}})$. A dish that contains structural starch (e.g., a salad with croutons) is reclassified as nachos (outer container plus inner starch), not salad. Accordingly, salad cells are generated and analyzed only at `starch_type=none`, while non-salad morphologies require a non-`none` starch type. Figure 2 summarizes the nine cube-type morphologies used on axis i.

3.1 The Rice Exception, Mono-Foods, and Dish-Level Classification

The Cube Rule (cuberule.com) explicitly declines to rule on one case: "You are free to interpret the nature of rice however you wish." It is the only food for which the Rule gives no cube ruling. The Rule *does* address rice in context: nigiri is toast, sushi rolls are sushi, ramen is nachos. The ambiguity is thus not rice-in-dishes but *plain rice by itself*—rice as a mono-food. The Rule's nachos examples (couscous, poutine, ramen) are illustrated with other things mixed in (couscous with add-ins, poutine with cheese and gravy); nachos is thus starch *plus* other components, not a bowl of plain starch alone. We therefore speculate that the exception applies to the nature of *plain rice alone*, and that a similar ambiguity holds for other mono-foods: a bowl of plain pita chips, a bowl of bread crumbs—single-starch items with no second component.

We propose that plain starches by themselves, as with mashed potatoes (which the Rule classifies as salad, i.e. soup as wet salad), are *salads*, and that there exists a gradient from salad to nachos with the addition of more things: lettuce with no croutons is salad; lettuce with croutons is nachos; but a bowl of croutons alone reverts to salad, as it is a single food. In analogy to the horseshoe theory of political economy—where the far left and far right are argued to resemble each other more than either resembles the center [1]—we suggest this be considered the horseshoe theory of salad: the extremes (all lettuce or all croutons) are both salad, while the middle (lettuce and croutons together) is nachos (Figure 3). Such cases may also warrant a new category entirely: foods that are a single starch, distinct from both "no structural starch" (salad) and "outer container plus inner

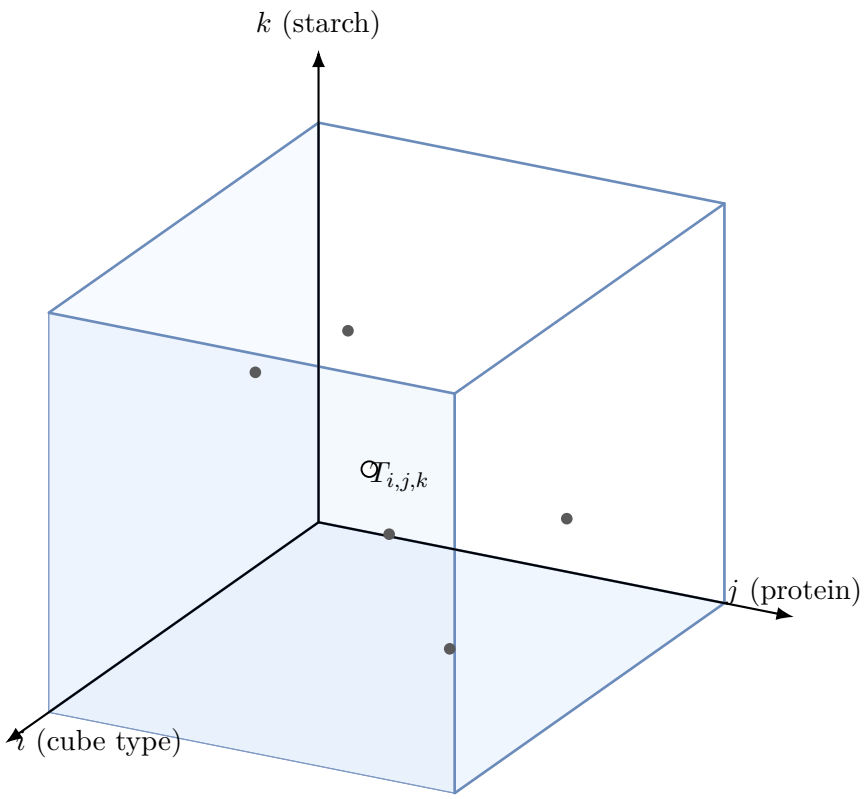

Figure 1: Tensor schematic. The cube-type, protein, and starch axes define the index space of T; filled markers illustrate occupied cells in $\mathcal{S}$, and the highlighted marker indicates a generic cell (i, j, k).

starch" (nachos). In this work we do not add that category; we classify at the *dish* level so that each dish receives one (i, j, k) from its structural starch placement, and we leave mono-starch boundary cases (plain rice, plain croutons) to the Rule's open interpretation.

3.2 Constrained LLM Candidate Generation

Cells $(i, j, k) \in \mathcal{E}$ (candidates in $\mathbb{F}_\infty \setminus \mathbb{F}_{\text{real}}$) are queried with a morphology-first prompt that includes a compact Cube Rule crash course, type-specific definitions, and canonical Cube Rule examples. This prompt can be viewed as a small prompt taxonomy: it separates structural morphology instructions, axis-definition instructions, and admissibility constraints so that each part of the classification problem is stated explicitly. The prompt enforces exact axis matching for cube morphology, protein class, and starch class rather than semantic similarity. For example, bottom-face-only starch placements are treated as toast morphology even when the dish name is not colloquially "toast". The API call uses schema-constrained generation rather than free-form text parsing: candidates are returned through a tool payload with typed fields (`shortname`, `description`, `is_real`, `confidence`, and `rationale_brief`). Operationally, this is also a practical form of grammar-constrained decoding, since only outputs that satisfy the required field structure can be returned. Prompt instructions also prioritize commonly recognized dish names over compositional labels; when only weak descriptive names are available, the model is instructed to return no candidate. The inclusion of canonical Cube Rule examples functions as lightweight retrieval-augmented generation: the model is not asked to classify from an unconstrained prior, but from an in-prompt

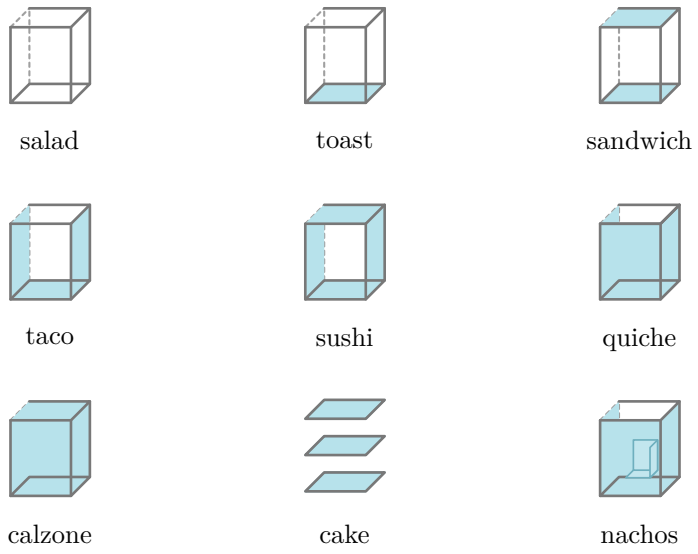

Figure 2: Cube-type morphology schematics for the nine categories on axis i. The blue regions indicate structural starch placement; the drawings are schematic rather than literal geometric reconstructions.

bank of morphology-defining exemplars. Prompt scaffolding also states that cube-type labels are morphology-only tags: for example, "salad" does not imply a cold dish, and "taco" does not imply spice or cuisine identity. Protein semantics are also made explicit in prompting and filtering: `none` is reserved for cases without meaningful protein contribution, while dairy ingredients (for example milk, cream, yogurt, and cheese) are treated as dairy-protein signals rather than `none`.

3.3 Model and Cost Disclosure

All LLM-assisted generation runs used Anthropic Claude (model: `claude-opus-4-6`). The total API cost for the full set of reported experiments was $5.77.

3.4 Hard Validity Constraints

The generation prompt includes hard constraints for strict matching:

- no starch substitution across classes (for example, pastry dough is not interchangeable with pita or other wheat bread forms),
- no protein substitution across classes,
- no composed "plate" outputs that combine separate foods (for example, "X served with Y"),
- conservative null behavior: if no obvious match exists, the model should return an empty candidate list.

Additionally, cube/starch pairings that violate the structural ontology are prohibited: salad is generated only in the `starch_type=none` slice, and non-salad morphologies are generated only with non-`none` starch types.

These same constraints are enforced after generation using deterministic lexical guardrails, yielding a lightweight constraint-satisfaction layer over candidate rows. In that sense, the pipeline

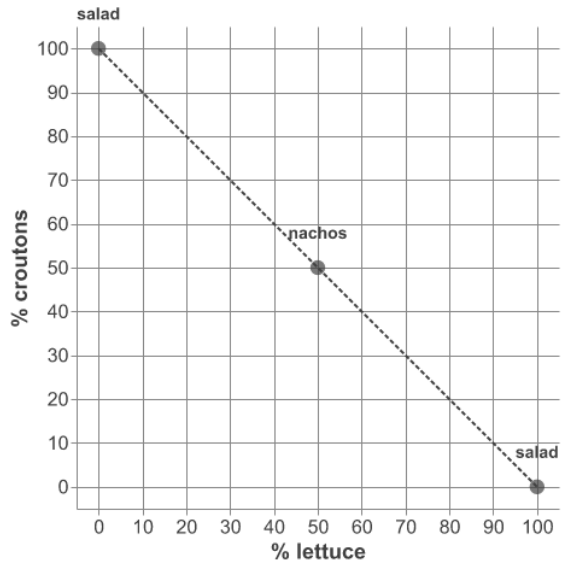

Figure 3: Horseshoe theory of salad: percent lettuce vs. percent croutons. The extremes $(0, 100)$ and $(100, 0)$ are salad (single component); the middle $(50, 50)$ is nachos. Dotted line: lettuce-crouton proportion gradient.

combines stochastic proposal with guided decoding: the model is steered both at generation time and at acceptance time toward outputs that satisfy the ontology. Candidates failing these checks are rejected before they enter the LLM candidate table for downstream review. Additionally, candidates below a configured confidence threshold are dropped so uncertain guesses do not set $T_{i,j,k} \leftarrow 1$ for speculative (i, j, k). In brief, candidate proposal is stochastic, but acceptance is governed by deterministic constraints and review.

3.5 Ground-Truth Calibration

We validate the filtering pipeline against a ground-truth set derived from canonical Cube Rule examples. The validation script checks axis mapping consistency and verifies that canonical Cube Rule examples pass the same lexical and protein/starch guardrails used for LLM-generated rows. During calibration, several false negatives were identified and corrected by refining token matching and conflict rules (for example, avoiding substring-only matches and reducing over-restrictive composition heuristics). After these adjustments, all canonical Cube Rule examples pass validation under the active guardrail configuration.

3.6 Visualization Axis Ordering

For occupancy visualizations, the order of indices on each axis (e.g., which i appears left-to-right) is computed with hierarchical clustering rather than fixed ontology order. Each index is represented by its binary occupancy profile (e.g., for fixed i, the vector $(T_{i,1,1}, \ldots, T_{i,J,K})$ flattened); pairwise dissimilarity is computed with Jaccard distance, and leaves are ordered by average-linkage dendrogram seriation. This ordering groups indices with similar occupancy structure and is used consistently in both pairwise projections (e.g., $T_{i,j,\cdot}$ over i, j) and 3D categorical axes, producing an interpretable occupancy manifold view of the tensor slices. At a descriptive level, this reflects a weak manifold hypothesis: foods that are nearby in ontology space tend to exhibit related occupancy

patterns rather than arbitrary dispersion across the tensor.

4 Experiments

The experiment workflow has three steps: map observed foods to coordinates (i, j, k) and set $T_{i,j,k} = 1$ accordingly; measure occupancy $|\mathcal{S}|$ and collisions (multiple foods at the same (i, j, k)); and generate candidate foods for cells in $\mathcal{E}$. The third step can be interpreted as a constrained categorical form of tensor completion: we do not impute arbitrary values, but instead propose named dishes for previously unoccupied cells under hard ontological constraints. Initial analysis reports which regions of the index set $\{1, \ldots, I\} \times \{1, \ldots, J\} \times \{1, \ldots, K\}$ are dense ($T_{i,j,k} = 1$ for many (i, j, k) in a neighborhood), sparse, or structurally inconsistent under the chosen ontology. From a representation-learning perspective, this is a small-scale study of combinatorial generalization under hard axis constraints on i, j, and k. Empirically, the workflow tests whether culinary naming conventions can be mapped consistently onto the proposed coordinate system. Axis categories in Figure 4 and Figure 5 are ordered by occupancy-profile hierarchical clustering as described in the Methods section (applied to the indices i, j, k along each axis).

Review outcomes. All LLM-generated candidates exported for review were accepted and included in the real-only tensor views; the distribution of model confidence for those accepted candidates is shown in Figure 6. These accepted rows can therefore be read as reviewed tensor-completion proposals rather than unconstrained generative samples.

We also run a targeted mini-experiment on foods whose names explicitly contain "salad" (for example, chicken salad, egg salad, Snickers salad[1], and jello salad[2]), asking the model to classify each item by structural starch placement and thus to assign a cube index i (and implicitly j, k). Table 1 reports only the subset that violates the Cube Rule salad definition (no structural starch faces), which in the tensor corresponds to the 'salad' morphology with `starch_type=none`. As a secondary aside experiment, we test the hypothesis that most violations arise from discrete starch inclusions (for example, croutons, pasta pieces, or candy chunks), which shift morphology from the salad index i_{salad} toward the Cube Rule "nachos" index i_{nachos} (outer dish plus inner starch).

5 Discussion

If the number of distinct foods we assign to coordinates exceeds IJK, then by the pigeonhole principle at least one cell (i, j, k) must contain more than one food; that is, collisions are guaranteed when $|\mathbb{F}_{\text{real}}| > IJK$ (or when mapping any subset of $\mathbb{F}_\infty$ of size $> IJK$ into the index space). This motivates optional extensions with additional axes (e.g., a fourth index ℓ), which can be interpreted as moving from $T \in \{0, 1\}^{I \times J \times K}$ toward a higher-order multiway representation. One practical tension is model granularity: under-specified ontologies (small I, J, K) collapse semantically distinct foods into the same (i, j, k), while over-specified ontologies create extreme sparsity ($|\mathcal{S}|/IJK$ very small) and weak empirical support per coordinate. Future extensions should resolve boundary cases where sugar-dominant processed inclusions (for example nougat/candy in Snickers

[1]Snickers salad is a dessert-style potluck dish, commonly made with chopped Snickers bars, tart apples, and whipped topping (often with pudding), and is widely associated with Upper Midwest potluck and church-supper traditions (especially Minnesota and Iowa), though no single point of origin is well documented.

[2]Jello salad is a molded gelatin dish, often containing fruit, marshmallows, or whipped topping, that appears in church-supper, holiday, and potluck traditions across the Upper Midwest and other parts of the United States; despite the name, its structure is far removed from the vegetable-forward sense of "salad."

Food	Salad?	Cube Type	Starch Faces	Rationale
potato salad	no	nachos	none	Discrete starch pieces (potato) mixed into a dressed base. Inner-starch pattern → nachos.
macaroni salad	no	nachos	none	Discrete pasta pieces mixed with dressing/veg. Inner-starch pattern → nachos.
pasta salad	no	nachos	none	Discrete pasta pieces mixed with vegetables and dressing. Inner-starch pattern → nachos.

Table 1: Mini-experiment: only foods containing the word salad were tested, and only violations of the Cube Rule salad definition (no structural starch faces; in the tensor, salad with `starch_type=none`) are reported.

salad) may trigger non-salad intuitions without mapping cleanly to the current starch index k. The Cube Rule's rice exception (plain rice as a mono-food) and the proposed gradient from salad to nachos (single starch → salad; starch plus other things → nachos) suggest a possible extension: a dedicated category for *single-starch* foods, distinct from salad and nachos, which would resolve the "interpret as you wish" for plain rice, plain croutons, and similar mono-foods without collapsing them into salad. Our current dish-level classification leaves that boundary to the Rule's open interpretation. Relatedly, jello-style dishes suggest that gelatin may warrant a separate protein index j beyond the current formalism, likely yielding a very sparse but interesting subset of $\mathcal{S}$ in the morphological state space (with soup dumplings as one motivating example, since they classify as calzone under Cube Rule morphology and include gelatinized broth). Topological language is useful here: regions where $T_{i,j,k} = 1$ is dense in index space behave like low-curvature neighborhoods, while isolated $(i, j, k) \in \mathcal{S}$ resemble high-curvature outliers in a discrete morphological state space. If local adjacency relations are imposed on occupied cells, $\mathcal{S}$ can also be viewed as inducing a sparse simplicial complex over the index lattice, with missing bridges and cavities corresponding to structurally absent combinations. Under that view, persistent homology and broader topological data analysis provide language for discussing connected components, holes, and other mesoscale features of the food-occupancy pattern as the ontology grows. Operationally, the ontology should avoid both excessive coarseness and excessive fragmentation: I, J, and K should be large enough to separate distinct dishes, but restrained enough to keep $|\mathcal{S}|$ empirically supportable.

6 Conclusion

We have shown that the Cube Rule of Food can be expressed as a geometric classification problem: dishes map to coordinates (i, j, k) in a third-order tensor T, with cube morphology i, protein taxonomy j, and starch substrate k as the axes of a discrete culinary state space. Canonical Cube Rule examples occupy well-defined cells; empty cells correspond to conceivable-but-unobserved dishes; and collisions—multiple foods at the same (i, j, k)—become topological facts rather than matters of taste. By constraining an LLM to propose candidates under strict axis matching, lexical guardrails, and a review gate, we have extended the support set $\mathcal{S}$ beyond the canonical seed while keeping the tensor falsifiable and the Rule's edge cases explicit: salad occupies the `starch_type=none` slice; the rice exception applies to plain rice as a mono-food; and the horseshoe theory of salad captures the gradient from single-component salad (all lettuce or all croutons) to nachos (lettuce and croutons together). Methodologically, this combines prompt taxonomies,

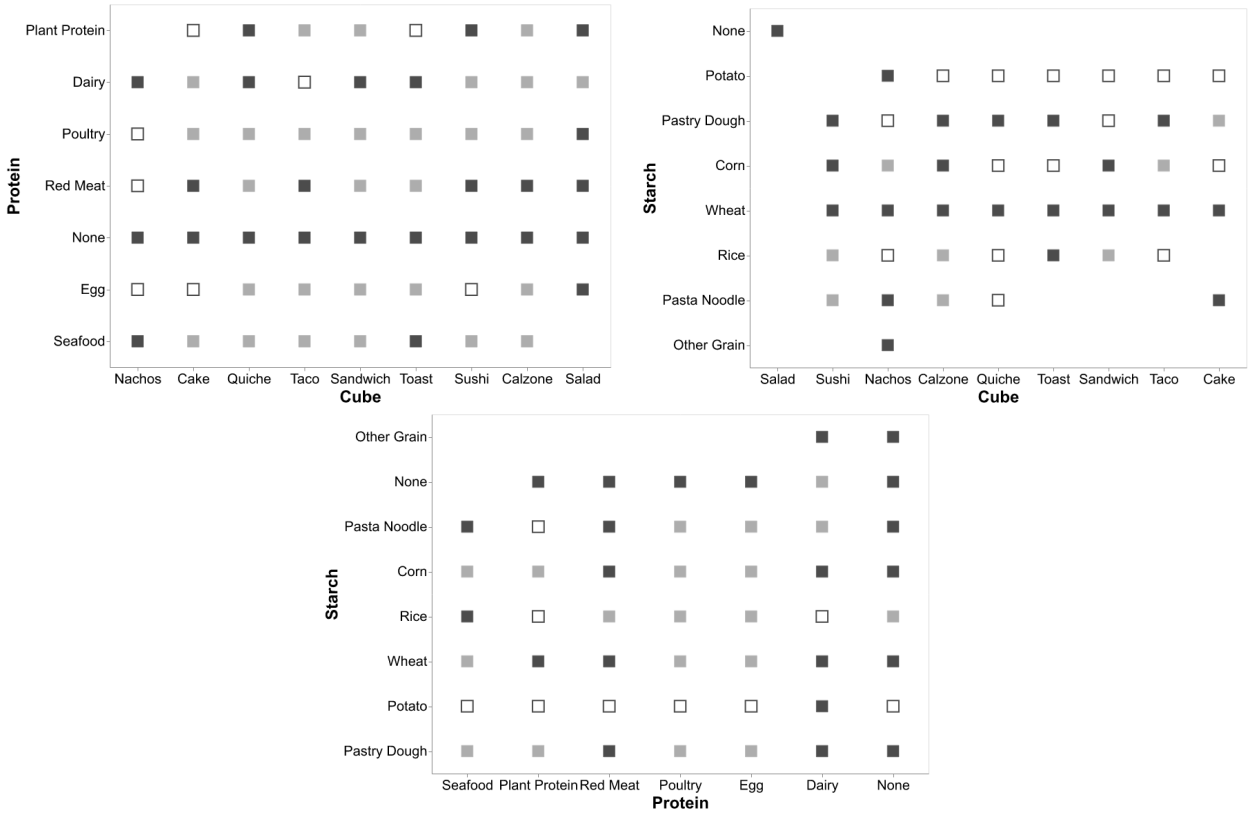

Figure 4: Pairwise projections of T: slices over (i, j), (i, k), and (j, k). Black squares mark canonical (i, j, k) with $T_{i,j,k} = 1$, gray squares mark accepted real LLM-identified cells, and white squares with black outlines mark LLM speculative cells; cells in $\mathcal{E}$ are shown without glyphs. Omitted categories by projection: Cube vs. Protein (Cube: none; Protein: none); Cube vs. Starch (Cube: none; Starch: none); Protein vs. Starch (Protein: none; Starch: none).

lightweight retrieval-augmented generation, guided decoding, and constrained tensor completion within a single food-classification pipeline.

This formalism can also be extended in principled ways. The index sets I, J, and K need not remain fixed. The cube axis i may gain a dedicated *single-starch* category, resolving the Rule's "interpret as you wish" for plain rice and bowls of croutons without collapsing them into salad. The protein axis j may admit gelatin (and thus jello, soup dumplings, and other gelatin-bound dishes) as a distinct type, yielding a sparse but coherent region of $\mathcal{S}$. The starch axis k may be extended to include sugar-dominant substrates—candy, nougat, Snickers bars—so that Snickers salad and its ilk cease to be boundary cases and take their place at a named coordinate. Each such expansion refines the representation of conceivable foods within the same indexing framework.

More broadly, this framework treats morphology, protein, and starch as formal indices rather than informal descriptors. Future work can therefore extend the same tensor formalism while keeping the convention that salad occupies the `starch_type=none` slice, the rice exception, and mono-food boundary cases explicit.

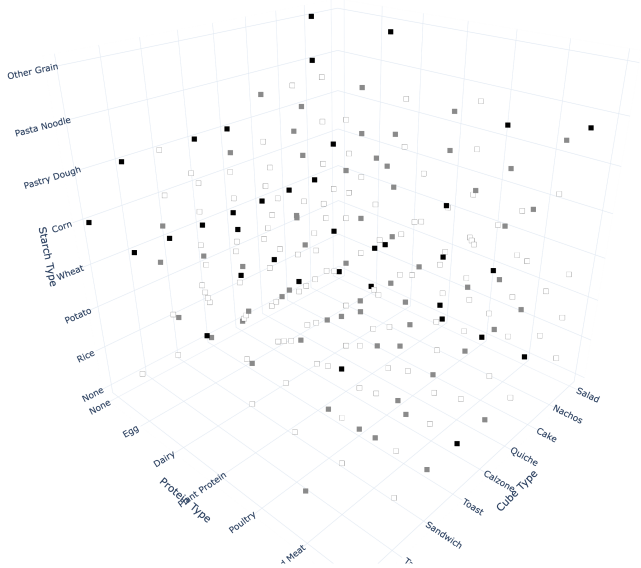

Figure 5: Static 2D rendering of the 3D occupancy tensor T. Black markers indicate canonical $(i, j, k) \in \mathcal{S}$, gray markers indicate accepted real LLM-identified cells, and white markers with black outlines indicate LLM speculative cells; $(i, j, k) \in \mathcal{E}$ are shown without glyphs.

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References

[1] Horseshoe theory. `https://en.wikipedia.org/wiki/Horseshoe_theory`.

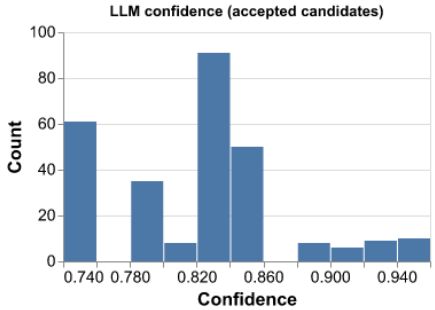

Figure 6: Distribution of model confidence for accepted LLM-generated candidates. The shape is visibly non-parametric because these confidence values are bounded self-reports on $[0, 1]$, then truncated by the acceptance cutoff at 0.75, so the observed mass piles up at a few high-confidence values rather than following a Gaussian law.